

Kodnatthaphat Authaianurak

ROYAL THAI GOVERNMENT (OCSC) SCHOLAR FULLY FUNDED MSC + PHD (2025-2030)

RESEARCH INTERESTS

- Artificial Intelligence & Machine Learning
- Computer Vision & Perception
- Robotics & Intelligent Systems
- Embedded Computing & IoT Architectures
- Edge/On-device AI
- Autonomous Decision-Making
- Environmental Sensing & Smart Systems

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Address: Adelaide, Australia

EDUCATION

Master of Computer Science (Artificial Intelligence) | 2025 -2026

Flinders University, Adelaide, Australia

- Expected Completion: Dec 2026 | Current GPA: 6.12/7.00
- Royal Thai Government (OCSC) Scholarship (Full Scholarship for MSc and PhD in AI and Robotics)

Bachelor of Engineering (Mechanical Engineering) | 2015-2018

Kasetsart University, Bangkok, Thailand

GPA: 3.10 / 4.00

Thesis: Weed Killing Robot – Development of a Mobile Robot for Spraying Herbicide Using Object Detection Techniques in a Cassava Farm

- Implemented Faster R-CNN Inception V2 for weed detection and autonomous spraying.
- Built a working prototype using Raspberry Pi, servo motors, and solenoid valves.
- Achieved 84.67% accuracy in real-time indoor trials.
- Supervisor: Asst. Prof. Hataitep Wongsuwarn

RESEARCH EXPERIENCE & SELECTED PROJECTS

Research Assistant | 2022-Present

Thailand Institute of Scientific and Technological Research (TISTR)

- Conduct research aligned with national strategic initiatives, applying AI and computer vision for automation and environmental monitoring.
- Coordinate multi-stakeholder projects, prepare research proposals, and present experimental results to government agencies.

Project 1: Pedestrian Crossing Detection Using Artificial Intelligence Control (2025)

Roles: Project Leader, Thailand Institute of Scientific and Technological Research (TISTR) (2024)

- Developed a YOLOv10-based smart pedestrian crossing for real-time human detection and PLC-based traffic control.
- Designed a compact computer vision pipeline and tested the system under outdoor lighting and weather variations.

Project 2: Intelligent Detection and Warning System for Forest Conservation and Wildlife Protection (2024)

Roles: Project Leader, Thailand Institute of Scientific and Technological Research (TISTR) (2024)

- Designed and implemented a LoRa-based IoT monitoring network using the Meshtastic framework for forest protection.
- Developed custom firmware and a web dashboard for real-time monitoring and community alert notifications.
- Improved wildlife safety by reducing escape incidents and enhancing early-warning capabilities in protected areas.

Project 3: Development of a Data Analytics Platform Prototype for PM2.5 Forecasting and Environmental Management (2023)

Roles: Project Member, Thailand Institute of Scientific and Technological Research (TISTR) (2024)

- Contributed to the concept design and prototype development of a data analytics platform for PM2.5 forecasting.
- Assisted in implementing the initial machine learning pipeline and dashboard mock-up for air quality visualization.
- Supported data collection and system testing for early-stage platform evaluation.
- The project aimed to provide data-driven tools for environmental pollution management and policy planning.

PROFESSIONAL EXPERIENCE

Process Improvement Engineer — Inoue Rubber (Thailand) PLC Mar 2019 – Jan 2022

- Applied lean manufacturing and root cause analysis to optimize production processes and tooling design.
- Created and standardized SOPs for process control and quality improvement.
- Collaborated with cross-functional teams to resolve production bottlenecks and enhance productivity.

Machine Vision Engineer — Robot System Co., Ltd. Apr 2018 – Mar 2019

- Integrated 3D vision cameras (Gocator) with robotic grinding systems for automated workpiece alignment.
- Implemented real-world coordinate calibration and robotic motion control for precision manufacturing.

SCHOLARSHIPS & AWARDS

- Royal Thai Government Scholarship (2025–2030): Full scholarship for Master's and PhD in Artificial Intelligence and Robotics.
- TGI x INNOMAC ACCEL Startup Grant (2018): 400,000 THB seed funding for commercializing the Weed Killing Robot project.

TECHNICAL SKILLS

AI & ML:

Python, TensorFlow, PyTorch (basic), Data Mining, Neural Networks

Computer Vision:

OpenCV, YOLO models, 3D Vision (Gocator), Object Detection Pipelines

Robotics & Automation:

NVIDIA Jetson, Basic ROS, PLC (Omron, Mitsubishi), Isaac sim

Embedded & IoT:

Arduino, ESP32, STM32, LoRa (Meshtastic), MQTT, Edge ML, Nvidia Jetson

Software Engineering:

FastAPI, Node-RED, Next.js, React, n8n, Linux/Ubuntu

Mechanical & Hardware:

Fusion 360, SolidWorks, 3D printing, sensor integration

LANGUAGES

- ENGLISH: IELTS 6.0 | PTE 57 (2024)
- THAI: NATIVE

REFERENCES

- Dr. Suwit Auchariyamet, Thailand Institute of Scientific and Technological Research (TISTR)
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- Dr. Natthapon Ardhan, Thailand Institute of Scientific and Technological Research (TISTR)
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